

Divide and Conquer recursion

1. For each application below, choose a vector or a linked list and defend your choice in two or three sentences. State any assumption your answer depends on.
 - (a) A simulation repeatedly looks up objects by integer index and occasionally adds an object at the end.
 - (b) An editor repeatedly removes the item at its current position and inserts another item there. It keeps a pointer to the current node.
 - (c) A program receives values one at a time, stores them, and later processes them in order. It does not know the number of values in advance.
 - (d) A program frequently inserts an item at a specified index near the middle.

For parts (c) and (d), is the choice as clear as it first appears? What information about actual usage would help you decide?